

Solutions to Toric Varieties by Cox, Little, Schenck

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Contents

1	Affine Toric Varieties	3
1.0	Background: Affine Varieties	3
1.1	Introduction to Affine Toric Varieties	6

1 Affine Toric Varieties

1.0 Background: Affine Varieties

Exercise 1.0.1

Suppose that V is an affine variety such that $R = \mathbb{C}[V] = \frac{\mathbb{C}[x_1, \dots, x_n]}{\mathbb{I}(V)}$. Then $\mathbb{I}(V)$ is a radical ideal so $\mathbb{C}[V]$ is finitely generated with no non-zero nilpotents.

Conversely, suppose that R is a finitely generated $\mathbb{C}$ -algebra with no non-zero nilpotents. By the universal property of finitely generated $\mathbb{C}$ -algebras, there exists an ideal $I \subseteq \mathbb{C}[x_1, \dots, x_n]$ such that $R \cong \frac{\mathbb{C}[x_1, \dots, x_n]}{I}$. Let $V = \mathbb{V}(I)$. Then $\mathbb{I}(V) = \sqrt{I}$ by the nullstellensatz and $\frac{\mathbb{C}[x_1, \dots, x_n]}{I} \cong \frac{\mathbb{C}[x_1, \dots, x_n]}{\sqrt{I}}$ so that R is the coordinate ring of V .

Exercise 1.0.2

- (a) Suppose that $a_1, a_2, c_1, c_2 \in R$ and $b_1, b_2, d_1, d_2 \in S$ such that $a_1/b_1 \sim a_2/b_2$ and $c_1/d_1 \sim c_2/d_2$. Then there exists $u, v \in S$ such that $u(a_1b_2 - a_2b_1) = 0 = v(c_1d_2 - c_2d_1)$. Then we have

$$\frac{a_1}{b_1} + \frac{c_1}{d_1} = \frac{a_1d_1 + b_1c_1}{b_1d_1} \quad \text{and} \quad \frac{a_2}{b_2} + \frac{c_2}{d_2} = \frac{a_2d_2 + b_2c_2}{b_2d_2}$$

Then notice that

$$(a_1d_1 + b_1c_1)(b_2d_2) - (a_2d_2 + b_2c_2)(b_1d_1) = (d_1d_2)(a_1b_2 - a_2b_1) + (b_1b_2)(c_1d_2 - c_2d_1)$$

Multiplying uv gives 0. Hence addition is well defined.

For multiplication, we have

$$\frac{a_1}{b_1} \frac{c_1}{d_1} = \frac{a_1c_1}{b_1d_1} \quad \text{and} \quad \frac{a_2}{b_2} \frac{c_2}{d_2} = \frac{a_2c_2}{b_2d_2}$$

Then we have

$$a_1c_1b_2d_2 - a_2c_2b_1d_1 = (c_1d_2)(a_1b_2 - a_2b_1) - (a_2b_1)(c_1d_2 - c_2d_1)$$

Multiplying uv gives 0. Hence multiplication is well defined.

- (b) Suppose that $r \in R$ and $s \in S$ such that $(r/s)^n = 0$ for some $n \in \mathbb{N}^+$. This means that there exists $u \in S$ such that $ur^n = 0$. Multiplying u^{n-1} on both sides give $(ur)^n = 0$. Since R has no nilpotents, we conclude that $ur = 0$. Then this implies $r/s = 0$. Thus $S^{-1}R$ has no nilpotents.

Exercise 1.0.3

- (a) By the above exercise, we know that R_f is a $\mathbb{C}$ -algebra. We now show that it is finitely generated. Suppose that R is finitely generated as a $\mathbb{C}$ -algebra by $r_1, \dots, r_n$. I claim that R_f is finitely generated as a $\mathbb{C}$ -algebra by $r_1, \dots, r_n, 1/f$. Let $r/f^k \in R_f$ be an arbitrary element. Then $r/f^k = 1/f^k \cdot r$ and r can be written as a sum and product of $r_1, \dots, r_n$ with coefficients. Hence we are done. By the above exercise, we know that if R has no nilpotents, then R_f has no nilpotents.
- (b) Recall from commutative algebra that there is a bijection

$$\text{Spec}(R_f) \xrightarrow{1:1} \left\{ m \subseteq R \mid \begin{array}{l} m \text{ is a maximal ideal} \\ \text{and } f \notin m \end{array} \right\}$$

The set on the right is the image of the continuous map $\text{Spec}(R_f) \rightarrow \text{Spec}(R)$ induced by the localization map $R \rightarrow R_f$. But the image is precisely the definition of $\text{Spec}(R)_f$.

- (c) Follows by definition of $\mathbb{C}(V)$ as the localization of $\mathbb{C}[V]$ at (0) and that the induced map $\mathbb{C}[V]_f \rightarrow \mathbb{C}(V)$ by the localization map is an injection.

Exercise 1.0.4

- (a) Clearly $1 \in S$ and if $f, g \in S$, then $f(p) \neq 0$ and $g(p) \neq 0$ implies $(fg)(p) \neq 0$ so that $fg \in S$.
 (b) We just need to show that the evaluation homomorphism $\text{ev}_p : \mathbb{C}[x_1, \dots, x_n] \rightarrow \mathbb{C}$ given by $f \mapsto f(p)$ for each $p \in V$ descends to $S^{-1}\mathbb{C}[V]$. Firstly, it descends to the quotient $\mathbb{C}[V]$ because if $f \in \mathbb{I}(V)$, then $f(p) = 0$ for all $p \in V$ by definition. To descend to $S^{-1}\mathbb{C}[V]$, note that if $f \in S$, then we have $\text{ev}_p(f) \in \mathbb{C}^\times$. Then by the universal property of localization, we conclude that the evaluation homomorphism descends to $S^{-1}\mathbb{C}[V]$.

I claim that $m_{V,p}$ is precisely the set of all non-units of $S^{-1}\mathbb{C}[V]$. Clearly if $f/g \in m_{V,p}$, then $f(p) = 0$ and $f \notin S$ so that f is not invertible. Now suppose that $f/g \notin m_{V,p}$. Then $f(p) \neq 0$ so that $f \in S$ and f is invertible. Since $m_{V,p}$ is the set of all non-units of $S^{-1}\mathbb{C}[V]$, we conclude that this is the unique maximal ideal.

- (c) Follows by definition of $\mathbb{C}(V)$ as the localization of $\mathbb{C}[V]$ at (0) and that the induced map $S^{-1}\mathbb{C}[V] \rightarrow \mathbb{C}(V)$ by the localization map is an injection.

Exercise 1.0.5

Let R be a UFD. Let $a/b \in \text{Frac}(R)$ that is integral. Assume that a, b do not have common factors. Then there exists $r_0, \dots, r_{n-1} \in R$ such that

$$\frac{a^n}{b^n} + r_{n-1} \frac{a^{n-1}}{b^{n-1}} + \dots + r_1 \frac{a}{b} + r_0 = 0$$

Rearranging, we get

$$a^n = -b(r_{n-1}a^{n-1} + \dots + r_1a + r_0b^{n-1})$$

This shows that any irreducible element dividing b also divides a^n , and hence a . Since a and b do not have common factors, this means that no irreducible element divides b . Since R is a UFD, b must be a unit. Hence $a/b \in R$.

Exercise 1.0.6

Notice that there is an isomorphism $\mathbb{C}[C] \cong \mathbb{C}[t^2, t^3]$ where $t = \bar{y}/\bar{x}$. Then the fraction field of C is given by $\mathbb{C}(C) = \mathbb{C}(t)$. We are given that t is integral. Hence $\mathbb{C}[t]$ is an integral extension of $\mathbb{C}[C]$ and that $\mathbb{C}[t] \subseteq \overline{\mathbb{C}[C]}$. Taking closure of $\mathbb{C}[C] \subseteq \mathbb{C}[t]$ gives $\overline{\mathbb{C}[C]} \subseteq \overline{\mathbb{C}[t]} = \mathbb{C}[t]$ so we are done.

Then the inclusion $\mathbb{C}[C] = \mathbb{C}[t^2, t^3] \hookrightarrow \mathbb{C}[t]$ induces the normalization map $\mathbb{C} \rightarrow C$ which is given by $t \mapsto (t^2, t^3)$.

Exercise 1.0.7

- (a) We want to show that $S^{-1}R$ is integrally closed in $\text{Frac}(R) = \text{Frac}(S^{-1}R)$. This means that we want to show $\overline{S^{-1}R} = S^{-1}R$. It is clear that $S^{-1}R \subseteq \overline{S^{-1}R}$. So let $g \in \overline{S^{-1}R}$. Suppose that $p(x) = x^n + \sum_{k=0}^{n-1} a_k x^k \in (S^{-1}R)[x]$ such that $p(g) = 0$. Choose $s \in S$ such that $sa_i \in R$ for $0 \leq i \leq n-1$. Then notice that $sg \in S^{-1}R$ satisfies the monic polynomial

$$q(x) = x^n + \sum_{k=0}^{n-1} s^{n-k} a_k x^k$$

since $q(sg) = s^n g^n + \sum_{k=0}^{n-1} s^n a_k x^k = s^n p(g) = 0$. But q is a polynomial in R since $s^{n-k} a_k \in R$. Thus we have that $sg \in \overline{R} = R$ since R is normal. This means that $g \in S^{-1}R$ and hence we conclude.

- (b) I claim that the intersection R is integrally closed in K . Let $x \in K$ be integral over R . Then $R \subseteq R_\alpha$ implies that x is integral over R_α . Since R_α is integrally closed, we conclude that $x \in R_\alpha$. Thus $\overline{R} \subseteq R$ so that R is integrally closed in K .

Now let $x \in \text{Frac}(R)$ be integrally closed in R . Since $\text{Frac}(R) \subseteq K$, we have $x \in K$ is integral over R so that $x \in R$. Hence $\overline{\text{Frac}(R)} = R$. Thus R is normal.

Exercise 1.0.8

Let $p \in \mathbb{C}^n$. I claim that $\frac{m_p}{m_p^2}$ has dimension n . Let $x_1 - p, \dots, x_n - p \in \mathbb{C}[x_1, \dots, x_n]$ be the coordinate functions. Clearly they lie in m_p . I claim that $(x_1 - p) + m_p^2, \dots, (x_n - p) + m_p^2$ is linearly independent in m_p/m_p^2 . Suppose that $\sum_{i=1}^n a_i((x_i - p) + m_p^2) = 0$ for some $a_i \in \mathbb{C}$. Then we have $\sum_{i=1}^n a_i(x_i - p) \in m_p^2$. But m_p^2 is generated by $x_i x_j$ for $1 \leq i, j \leq n$. Hence $a_1 = \dots = a_n = 0$. Hence $\dim_{\mathbb{C}}(m_p/m_p^2) = n$ and $\dim(T_p \mathbb{C}^n) = n$.

Exercise 1.0.9

By lemma 1.0.6, we conclude that $T_p V$ is isomorphic to $\text{coker}(d_p(f_1), \dots, d_p(f_s)) = \text{coker}(J_p(f_1, \dots, f_s))$. Then we have

$$\dim(T_p V) = n - \text{rank}(J_p(f_1, \dots, f_s))$$

So that V is smooth at p if and only if $\dim(T_p(V)) = d$ if and only if $\text{rank}(J_p) = n - d$.

Exercise 1.0.10

(a) Let $I = 0$. By the third bullet point, we have

$$0 = \bigcap_{l=1}^{\infty} m_{V,p}^l$$

Let φ be the map from $\mathcal{O}_{V,p}$ to $\mathbb{C}[[x_1, \dots, x_n]]$. Let $f \in \ker(\varphi)$. Then we have that $f + m_{V,p}^l$ corresponds to $\varphi(f) + m^l = m^l$ in the isomorphism in the second bullet point. Hence $\varphi(f) \in m^l$ so that $f \in m_{V,p}^l$. Hence $f \in \bigcap_{l=1}^{\infty} m_{V,p}^l$ so that $f = 0$. Hence φ is injective.

(b) Since b divides a in the power series ring, there exists $u \in \mathbb{C}[[x_1, \dots, x_n]]$ such that $\varphi(a) = \varphi(b)u$. For each $l \in \mathbb{N}^+$, there exists a unique $c \in \mathcal{O}_{V,p}$ such that $c + m_{V,p}^l$ corresponds to the elements $u + m^l$ using the isomorphism in the second bullet point. Now we have

$$\varphi(a) - \varphi(b)\varphi(c) = \varphi(a) - \varphi(b)u + \varphi(b)u - \varphi(b)\varphi(c) = \varphi(b)(u - \varphi(c)) \in \varphi(b)m^l \subseteq m^l$$

Hence $a - bc \in m_{V,p}^l$ and that $a \in (b) + m_{V,p}^l$. Using the third bullet point, we have that

$$a \in \bigcap_{l=1}^{\infty} (b) + m_{V,p}^l = (b)$$

so that b is divisible in a in $\mathcal{O}_{V,p}$.

(c) Notice that since $\mathcal{O}_{V,p}$ is a subring of $\mathbb{C}[[x_1, \dots, x_n]]$. Let $\tilde{\varphi} : \text{Frac}(\mathcal{O}_{V,p}) \rightarrow \text{Frac}(\mathbb{C}[[x_1, \dots, x_n]])$ be the injection induced by φ . Let $\alpha \in \text{Frac}(\mathcal{O}_{V,p})$ be integral. Then $\tilde{\varphi}(\alpha) \in \text{Frac}(\mathbb{C}[[x_1, \dots, x_n]])$ is integral. Since $\mathbb{C}[[x_1, \dots, x_n]]$ is normal, we conclude that $\tilde{\varphi} \in \mathbb{C}[[x_1, \dots, x_n]]$. Suppose that $\alpha = \frac{f}{g}$ for some $f, g \in \mathcal{O}_{V,p}$ and $g \neq 0$. Then we have $\varphi(g)\tilde{\varphi}(\alpha) = \varphi(f) \in \mathbb{C}[[x_1, \dots, x_n]]$ so that $\varphi(f)$ is divisible by $\varphi(g)$. By part (b), we conclude that f is divisible by g in $\mathcal{O}_{V,p}$. Hence $\alpha \in \mathcal{O}_{V,p}$. Thus $\mathcal{O}_{V,p}$ is normal.

Exercise 1.0.11

We already have $S \times W \subseteq \overline{S} \times W$ so taking closure gives $\overline{S \times W} \subseteq \overline{\overline{S} \times W} = \overline{S} \times W$ since $\overline{S} \times W$ is an affine variety.

Conversely, suppose that $\mathbb{I}(\overline{S \times W}) = (f_1, \dots, f_k) \in \mathbb{C}[[x_1, \dots, x_n, y_1, \dots, y_m]]$. Fix $w_0 \in W$. Then for each $s \in S$, we have $f_i(s, w_0) = 0$ so that $s \in Z_{w_0} = \mathbb{V}(f_1(-, w_0), \dots, f_k(-, w_0))$. Since Z_{w_0} is closed in S , we have $\overline{S} \subseteq Z_{w_0}$. Since w_0 is arbitrary, we conclude that $\overline{S} \times W \subseteq \overline{S \times W}$.

Exercise 1.0.12

Suppose that $V \times W = Z_1 \cup Z_2$ where Z_1, Z_2 are closed. Let $V_i = \{p \in V \mid \{p\} \times W \subseteq Z_i\}$. Then clearly we have $V_1 \cup V_2 \subseteq V$. On the other hand, suppose that $p \in V$. Since $\{p\} \times W$ is closed, we

must have $\{p\} \times W \subseteq Z_1$ or Z_2 . Hence $\{p\} \times W$ lies in Z_1 or Z_2 . Hence $V = V_1 \cup V_2$.

I claim that V_i is closed. For each $w \in W$. Let $\iota_w : V \rightarrow V \times W$ be the inclusion map. Since $\text{im}(\iota) = V \times \{w\}$, we have that ι_w is an isomorphism onto a closed subset. Then $\iota_w^{-1}(Z_i)$ is closed. Since

$$V_i = \bigcap_{w \in W} \iota_w^{-1}(Z_i)$$

we conclude that V_i is closed.

Since V is irreducible, we must have either $V = V_i$ for some i . In this case we have $V \times W = \bigcup_{p \in V} \{p\} \times W \subseteq Z_i$. Hence $Z_i = V \times W$.

1.1 Introduction to Affine Toric Varieties

Exercise 1.1.1

- (a) Let $I = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d) \subseteq \mathbb{C}[x_1, \dots, x_{d+1}]$. Clearly we have $\text{im}(\phi) \subseteq \mathbb{V}(I)$. For the reverse inclusion, notice first that if $d = 0$ then the result is trivial. Now let $p = (p_1, \dots, p_{d+1}) \in \mathbb{V}(I)$. If $p = 0$ then clearly $p \in \text{im}(\phi)$. Now let $m = \min\{k \mid p_k \neq 0\}$. I claim that $m = 1$ or $d + 1$. Notice that we have $x_{k-1} x_{k+1} - x_k^2 \in I$ for $2 \leq k \leq d$. If $2 \leq m \leq d$, then $p_{m-1} = 0$ so that $p_m = 0$, contradicting the minimality of m . Hence $m = 1$ or $d + 1$. If $m = d + 1$, then $p = (0, \dots, 0, p_{d+1})$. Since $p_{d+1} \in \mathbb{C}^\times$, choose $t \in \mathbb{C}$ such that $t^d = p_{d+1}$. Then we have that $(0, t) \mapsto p$ so $p \in \text{im}(\phi)$. If $m = 1$, let $r = p_2/p_1$. Using the relation $x_1 x_{j+1} = x_2 x_j$ we deduce that $p_{j+1}/p_j = r$. Hence p is given by $p = (p_1, p_1 r, \dots, p_1 r^d)$. Since $p_1 \in \mathbb{C}^\times$, choose $s \in \mathbb{C}$ such that $s^d = p_1$. Then we have $(s, rs) \mapsto p$ so that $p \in \text{im}(\phi)$. Thus we are done.
- (b) Since I is homogeneous, we know that $\sqrt{I}$ is homogeneous. Then $\mathbb{I}(\widehat{C}_d) = \mathbb{I}(\mathbb{V}(I)) = \sqrt{I}$ is homogeneous.

Exercise 1.1.2

Exercise 1.1.3

Exercise 1.1.4

- (a) First notice that $\mathbb{V}(\mathbb{I}(\text{im}(\phi))) = \overline{\text{im}(\phi)} = Y$ so applying $\mathbb{I}(-)$ on both sides give $\mathbb{I}(\text{im}(\phi)) = \mathbb{I}(Y)$. Let $h \in \mathbb{I}(\text{im}(\phi))$. Then we have $h(f_1(p), \dots, f_s(p)) = 0$ for all $p \in V$. Hence $h \circ (f_1, \dots, f_s) \in \mathbb{I}(V)$ so that $h \in \ker(\phi^*)$. Conversely, suppose that $g \in \ker(\phi^*)$. Then we have $g \circ (f_1, \dots, f_s) \in \mathbb{I}(V)$ so that $g(f_1(p), \dots, f_s(p)) = 0$ for all $p \in V$. Hence $g \in \mathbb{I}(\text{im}(\phi))$.
- (b) Use this to prove that $\mathbb{I}(Y_{\mathcal{A}}) = \mathbb{I}(\text{im}(\Phi_{\mathcal{A}})) = \ker(\pi) = \ker(\Phi_{\mathcal{A}})$.

Exercise 1.1.5

The claim is that $\mathbb{N}\{m_1, m_2, m_3, m_4\} = C \cap \mathbb{Z}^3$. Clearly we have $\mathbb{N}\{m_1, m_2, m_3, m_4\} \subseteq C \cap \mathbb{Z}^3$. Now suppose that $(x, y, z) \in C \cap \mathbb{Z}^3$.

I claim that $\max\{0, -z\} \leq \min\{x, y\}$. Suppose that $(x, y, z) = am_1 + bm_2 + cm_3 + dm_4$, then we have $0 \leq a = x - d$ and $0 \leq b = y - d$ and $d \geq -z$. Hence $-z \leq d \leq x, y$.

Let $\lambda_4 = \max\{0, -z\}$. Let $\lambda_1 = x - \lambda_4$, $\lambda_2 = y - \lambda_4$, $\lambda_3 = z + \lambda_4$. I claim that $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{N}$. Firstly λ_4 is trivially a non-negative integer. Since $\max\{0, -z\} \leq \min\{x, y\}$, we conclude that $\lambda_1, \lambda_2, \lambda_3 \geq 0$. Hence $(x, y, z) = \sum_{i=1}^4 \lambda_i m_i$. Thus we are done.

Exercise 1.1.6

- (a) I claim that Φ_1 is dominant. The affine semigroup corresponding to the affine toric variety of Φ_1 is generated by s^2, st, st^3 . Hence $\mathbb{C}[\overline{\text{im}(\Phi_1)}] = \mathbb{C}[s^2, st, st^3]$ and

$$\begin{aligned}\mathbb{C}(\overline{\text{im}(\Phi_1)}) &= \mathbb{C}(s^2, st, st^3) \\ &= \mathbb{C}(s^2, st, t^2) \\ &= \mathbb{C}(s^2, st, s^2t^2) \\ &= \mathbb{C}(s^2, st)\end{aligned}$$

Now $\mathbb{C}(s, t)$ is an algebraic extension of $\mathbb{C}(s^2, st)$ since s satisfies the polynomial $x^2 - s^2 \in \mathbb{C}(s^2, st)[x]$ and $t = (st)/s$. The transcendence degree of $\mathbb{C}(s, t)$ is 2 so that $\dim(\overline{\text{im}(\Phi_1)}) = 2$. Since $\mathbb{V}(xz - y^3)$ is irreducible of dimension at most 2 and $\overline{\text{im}(\Phi_1)} \subseteq \mathbb{V}(xz - y^3)$, we conclude that $\overline{\text{im}(\Phi_1)} = \mathbb{V}(xz - y^3)$. The proof is similar for Φ_2 .

- (b) For Φ_1 , notice that the point $(0, 0, 1)$ does not lie in the image because if $(s^2, st, st^3) = (0, 0, 1)$, then we have $s = 0$ and $t = 0$ and $st^3 = 1$, a contradiction. Hence Φ_1 is not surjective.

Let $p = (p_1, p_2, p_3) \in \mathbb{V}(xz - y^3)$. Suppose that $p_2 \neq 0$. Then $p_1, p_3 \neq 0$. Choose $s, t \in \mathbb{C}$ such that $s^3 = p_1$ and $t^3 = p_3$. Then we have $\Phi_2(s, t) = p$. Now suppose that $p_2 = 0$. Then $p_1 = 0$ or $p_3 = 0$. If $p_1 = p_3 = 0$, then we have $\Phi_2(0, 0) = (0, 0, 0)$. If $p_1 = 0$ and $p_3 \neq 0$, then choose $t \in \mathbb{C}$ such that $t^3 = p_3$. Then we have $\Phi_2(0, t) = p$. Finally, if $p_1 \neq 0$ and $p_3 = 0$, then choose $s \in \mathbb{C}$ such that $s^3 = p_1$. Then $\Phi_2(s, 0) = p$. Thus Φ_2 is surjective.

Exercise 1.1.7**Exercise 1.1.8**